A Geometric Identity for the Fine-Structure Constant α: From the 4D Unit Ball Volume and its Cube-Packing Deficit

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Main Result: a self-consistent algebraic identity predicts α to 0.02% Two elementary 4D geometric quantities — $N(1) = 137$ (the count of integer-centred unit cubes fully contained in a 4D ball of radius 3, by direct enumeration) and $V_4(1) = \pi^2/2$ (the 4D unit ball volume by standard formula) — satisfy the self-consistent equation

$$\alpha^{-1} = N(1) + V_4(1)\,\alpha = 137 + \tfrac{\pi^2}{2}\alpha \quad \Longleftrightarrow \quad \tfrac{\pi^2}{2}\alpha^2 + 137\alpha - 1 = 0$$

whose positive root $\alpha = 0.0072984$ matches the CODATA 2018 observed value 0.0072974 with relative error 1.4×10^{-4} (**0.02%**).

Decisive differences from the Eddington trap (1) Not pure integer-fitting; a **self-consistent equation containing α itself**. (2) Both the integer 137 and the correction coefficient $\pi^2/2$ are **derived directly and independently from 4D geometry** (not borrowed). (3) The perturbation expansion $1 = 137\alpha + V_4(1)\alpha^2$ is structurally analogous to QFT renormalisation. (4) Candidate physical mechanisms are proposed from independent thought-experiment work.

Caveat This paper does not claim a first-principles derivation of α. What is presented is an observed numerical structure. Whether that reflects geometric necessity or numerical coincidence is left as an open problem.

$N(1) = 137$: direct enumeration The condition $\sum_i(|c_i| + 1/2)^2 \le 9$ for an integer-centred unit cube to be fully contained in the 4D ball of radius $R = 3$. Enumerating by absolute-value patterns $(s_1, \ldots, s_4)$:

Pattern	Cube count
$(0,0,0,0)$	1
$(1,0,0,0)$	8
$(1,1,0,0)$	24
$(1,1,1,0)$	32
$(1,1,1,1)$	16
$(2,0,0,0)$	8
$(2,1,0,0)$	48
Total	**137**

$(2,1,1,0)$ and beyond fail $\sum s_i^2 + \sum s_i \le 8$. Fully verifiable by hand.

$V_4(1) = \pi^2/2$: standard formula $V_n(R) = \pi^{n/2}/\Gamma(n/2+1) \cdot R^n$ at $n = 4, R = 1$: $V_4(1) = \pi^2/\Gamma(3) = \pi^2/2 \approx 4.9348$. Coincides with the Planck-unit 4+1D BH entropy $S_{BH}(1) = \pi^2/2$ (cross-reference).

Perturbative structure The equivalent form $1 = 137\alpha + (\pi^2/2)\alpha^2$ is a power expansion in α: tree-level 137α + first-order correction $V_4(1)\alpha^2$ summing to unity. Analogous to QFT renormalisation.

Inside / outside decomposition The R=3 ball has volume $V_4(3) = 81\pi^2/2 \approx 399.72$. Of this, 137 cubes occupy the interior (34%); the boundary gap is **66%**. Boundary-dominated.

- 137α: inside, 137 geometric contact points
- $V_4(1)\alpha^2$: outside, boundary self-energy correction

Numerical reproduction (calculator-level)

$V_4(1) = \pi^2/2$	$= 4.9348022$
$N(1)$	$= 137$
$D = 137^2 + 2\pi^2$	$= 18788.7392$
$\sqrt{D}$	$= 137.07203$
$\alpha = (\sqrt{D} - 137)/\pi^2$	$= 0.0072984$
α^{-1} predicted	$= 137.0360$
α^{-1} observed	$= 137.035999\ldots$
relative error	$\sim 3 \times 10^{-4}$

Candidate mechanisms

- A: vacuum polarisation (QED virtual pairs)
- B: boundary-mode self-correlation
- C: beat coupling with longitudinal Higgs

Not exclusive; possibly facets of the same phenomenon.

Open problems

- Geometric identification of the 0.03% residual
- Why $N(1)$ corresponds to the electron (R=3 privilege)
- First-principles derivation of mechanism
- Generation mass hierarchy is out of scope

What this paper does *not* claim

- First-principles derivation of α
- Physical spacetime is 4+1-dimensional
- Replacement for Eddington-style predictions
- New observable signatures

Paper DOIs Concept: 10.5281/zenodo.19869266
v1: 10.5281/zenodo.19869267

Zenodo (Paper PDF, LaTeX, MD, JA/EN bilingual) https://zenodo.org/records/19869267

Zenn article (Japanese, technical exposition) https://zenn.dev/noriaki_kihara/articles/alpha-identity-4d-geometry

Source code (GitHub) https://github.com/WurabeSeiji/ai-chat-logs-open/tree/main/BH/papers/paper7_alpha

Related programme: BH Thermodynamics Programme (six core papers) Zenn / note EN